

LeetCode 2582: Pass the Pillow

FAANG INTERVIEW QUESTION

Problem Understanding & Timeline

Example Scenario based on handwritten notes: $n = 4$ people, time $t = 5$

Pillow Passing Timeline (0 to 5 seconds):

$t=0$: Person 1

$t=1$: Person 2

$t=2$: Person 3

$t=3$: Person 4 (End reached, direction reverses)

$t=4$: Person 3

$t=5$: Person 2 → Output is 2

Note: People are 1-indexed.

Logic & Intuition

- The direction of the pillow reverses every time it reaches the end of the line (either person 1 or person n).
- Each full directional pass (which we can call a **cycle**) takes $(n - 1)$ seconds.
- Therefore, the total number of completed cycles is $t / (n - 1)$.
- The remaining time (seconds left) in the current cycle is $rem = t \% (n - 1)$.
- **Handling Directions (Even vs. Odd Cycles):**
 - If the cycle count is **even**, the pillow is moving forward. We start passing from the *first* person. The final position is $rem + 1$.
 - If the cycle count is **odd**, the pillow is moving backward. We start passing from the *last* person backwards. The final position is $n - rem$.

C++ Solution

```
class Solution {
public:
    int passThePillow(int n, int t) {
        int cycles = t / (n - 1);
        int rem = t % (n - 1);

        if (cycles % 2 == 1) {
            return n - rem;
        }

        return rem + 1;
    }
};
```